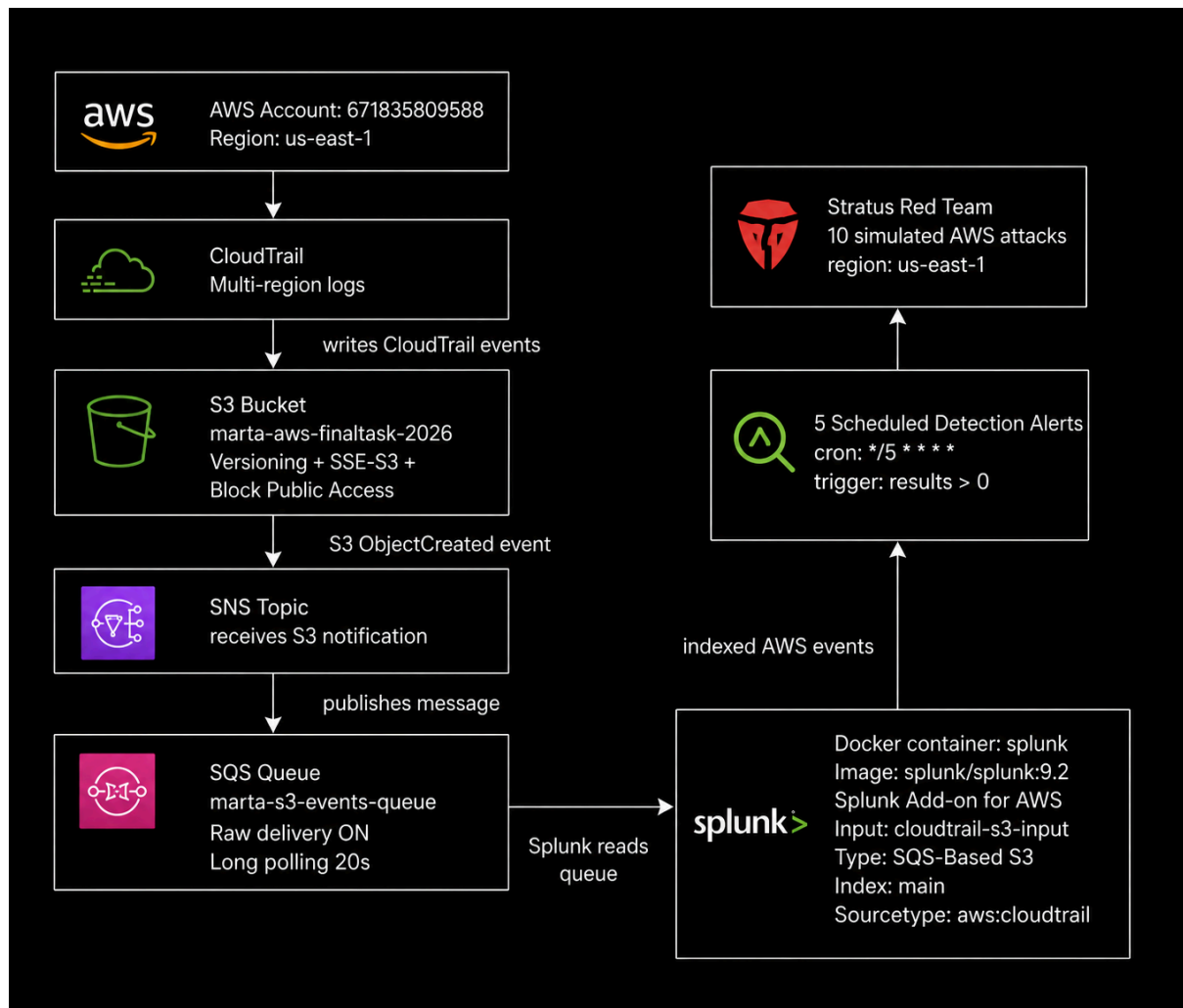




CloudTrail collects audit events within the AWS account. It is configured as a multi-region trail with management events enabled for both read and write operations. This allows monitoring of user and service activity across all AWS regions.

The S3 bucket marta-aws-finaltask-2026 is used to store logs. Versioning, SSE-S3 encryption, and Block Public Access are enabled to ensure log protection and retention.

The SNS Topic receives notifications about newly created log files in S3 through S3 Event Notifications. This allows new events to be automatically forwarded through the pipeline.

The SQS queue marta-s3-events-queue receives messages from SNS. Raw Message Delivery and Long Polling are enabled. The queue acts as a buffer between AWS and Splunk.

Splunk Enterprise is deployed locally in a Docker container. It receives AWS events, indexes them, and is used for monitoring and attack detection.

The Splunk Add-on for AWS provides integration with AWS services. Through the SQS-Based S3 Input cloudtrail-s3-input, CloudTrail events are ingested into the main index with the sourcetype aws:cloudtrail.

Five detection rules were created in Splunk to identify suspicious activity. They monitor CloudTrail modifications, IAM activity, secret access attempts, and potential data exfiltration behavior.

Stratus Red Team was used to simulate AWS attacks. A total of 10 TTPs were executed, generating CloudTrail events that were used to validate the detection rules and the overall monitoring pipeline.

Name	ATT&CK Tactic	Stratus TTPs Caught	Threshold Reasoning
CloudTrail Tampering	TA0005 Defense Evasion	cloudtrail-stop, cloudtrail-delete, cloudtrail-modify-event-selectors	Alert triggers when at least one CloudTrail modification event is detected because such actions are uncommon and highly suspicious in normal operations.
Logging and Monitoring Disabled	TA0005 Defense Evasion	cloudtrail-stop, vpc-flowlogs-delete	Threshold is set to greater than zero because disabling logging services directly impacts visibility and should always be investigated.
IAM Persistence Activity	TA0003 Persistence	iam-create-user, iam-create-access-key, iam-create-login-profile, iam-attach-user-policy	Creation of new IAM identities or credentials is relatively rare and may indicate persistence or privilege escalation attempts.
Credential and Secret Access	TA0006 Credential Access	secretsmanager-get-secret-value, ssm-get-parameter	Access to sensitive secrets generated by Stratus Red Team is considered suspicious

			and should trigger immediate review.
Snapshot Exfiltration and Risky Access Exposure	TA0010 Exfiltration	ebs-share-snapshot, iam-create-access-key , iam-attach-user-policy	Snapshot sharing and creation of additional access mechanisms may indicate preparation for data exfiltration or unauthorized access.

The most challenging part of the project was configuring the integration between S3, SNS, SQS and Splunk. Small configuration mistakes, especially permissions and notification settings, could break the entire pipeline. The most interesting aspect was observing how simulated attacks generated real CloudTrail events and how those events were detected automatically by the Splunk rules. If I repeated this project, I would add more AWS log sources and create a larger set of detection rules to improve ATT&CK coverage.